

Video Transcript

Video 9: API Documentation and Swagger (06:10)

Being able to fire functions on a remote web server is very powerful. Basically, we're doing distributed computing. We're firing functions on a remote machine, and we're doing it using HTTP. So, `http://localhost:5000` goes to the default route. If we add slash books, it fires the GET books function. If we go to slash add books, we'll go to an add book function. Now, this style of computing was formalized by Roy Fielding, a researcher, and he called it REST. This style of interfacing through a URL and calling a function, it stands for representational state interface and it's very popular for doing distributed computing. Now, one of the challenges we have is we need to know what data the particular API needs. So, for example, if we hit add books, then we need to provide data about the books. But the question is, what's the form of that data? Is it in the header? Is it in the body? Is it in form data? Is it in Json? So, we need to know the exact application interface and a company called Swagger has defined a method for defining these APIs. Let's take a look at Swagger.

So, here's the Swagger platform. If you just go to `swagger.io` slash tools you'll find documentation of the Swagger interface. Let me show you what a Swagger interface looks like. So, here for example, we're hitting, let me bring this down, we're hitting here localhost 5000. But now we're hitting slash swagger and it shows us here the interface that we have. So, here I've detailed, we have a login, we have books and we have addbooks. So, here we can see exactly the data that login needs, here it needs username and password. We can try it out. So, we can put in say test user, test user and execute it and we'll see what it does. It creates a curl request out of that data that we put in. So, here in the data, we'll see username equals test user, password equals test user, and the URL is slash login. And then it shows us the response body that's coming back. So, here we have basically the web page that we would get returned normally to us when we hit slash login. You can play with these interfaces. Let me just reload that. So, for example, you can see if you can add a book, but let's look at how this is defined in code.

So, let's go to our project. So, here we are. Now, there's various ways of doing this. I've chosen to use Json, so my file is `Swagger.json`, and it's under static here. And if we go to our API, we'll see

that we need to add `flask_swagger_ui`, and we'll import `Swaggerui_blueprint`. So, remember, if it's pip install, you do Flask dash, not underscore, Swagger dash ui. Now, we need to define some Swagger defaults. So, we're saying this is the URL, it'll be slash swagger. And we already hit that.

This is where we're going to define the Swagger is in `Swagger.json` and the static, the folder static. And then this is just some other configuration that you need to include. And finally, we register blueprint with our app and that's all we need to do to create that interface. Now, the Swagger file itself, as I said, it's in Json. You can also define it in YAML. I prefer Jason because it's easier to debug. YAML, like Python, is just based on offsets, so they don't have the curly braces. I find the curly braces are easier to check. Now, to define the paths, it's quite extensive. We have to put in a fair amount of data saying about what parameters, here we've got the login one, we need to save the parameter, we need username and password, and these are going to be strings.

When you use Swagger, you're going to have to look up in the Swagger documentation exactly how to lay out this file. In our post, we're doing a multi part form and in this case we're inputting a book and it has an author and a title and both of those are strings. So, you can get used to the Swagger structure fairly easily. So, you should know about Swagger, how it's implemented, and how to use it. In this course, we won't ask you to learn this, but you can certainly play with this and try adding a root for yourself or changing something about one of the roots. So, that's Swagger and you'll find all the documentation up there homepage. Bye for now.